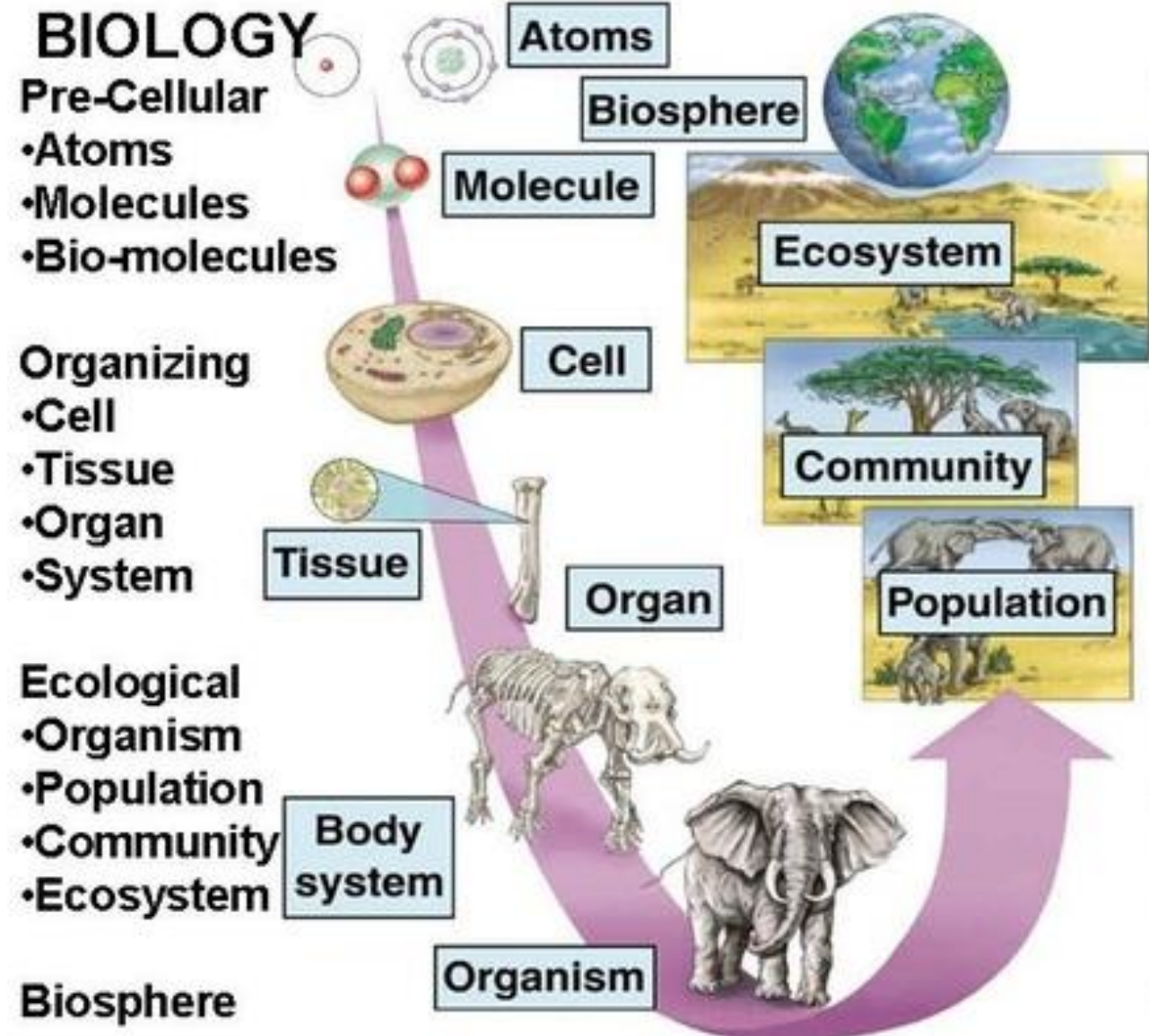


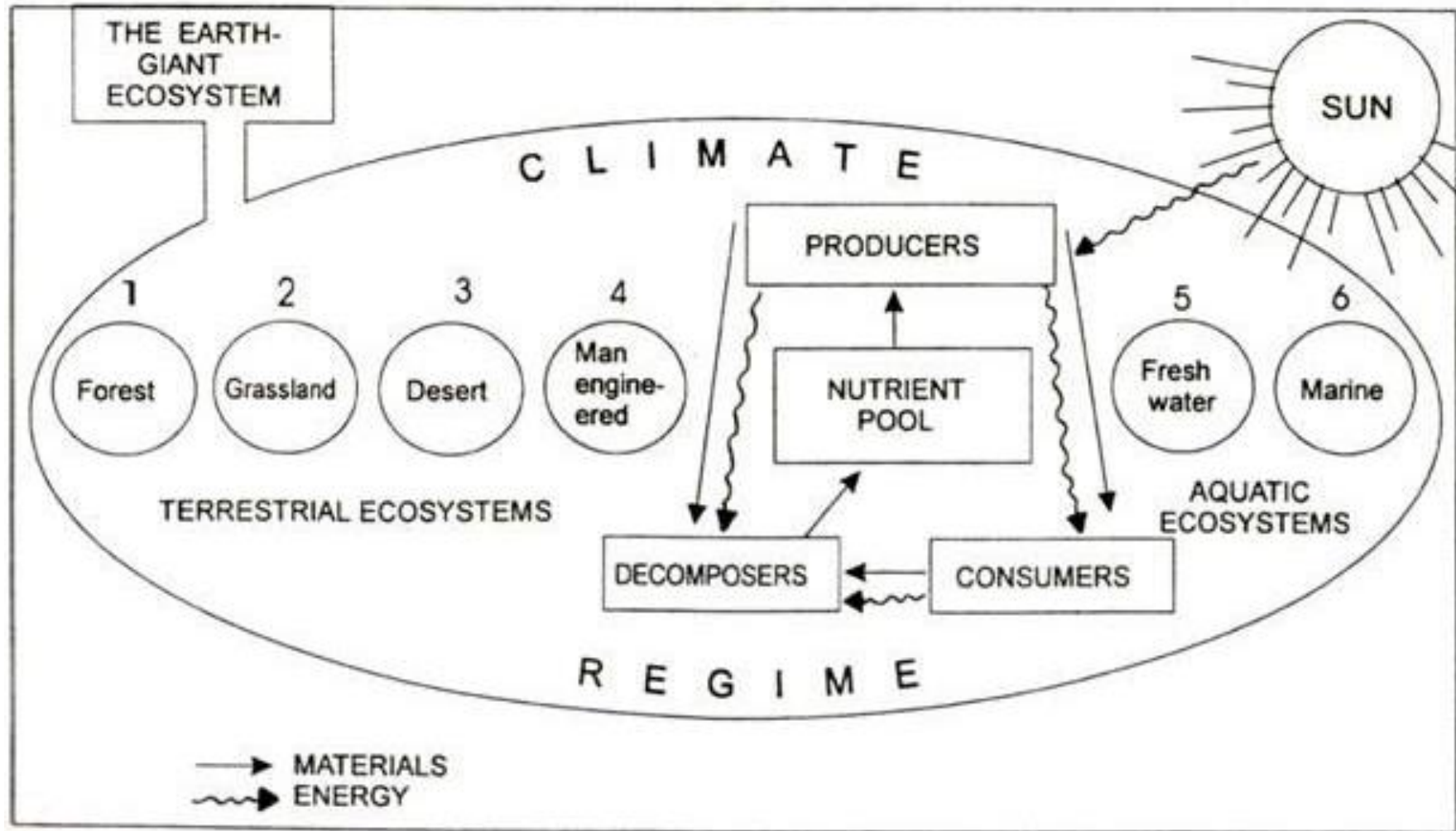


Ecology and Ecosystem

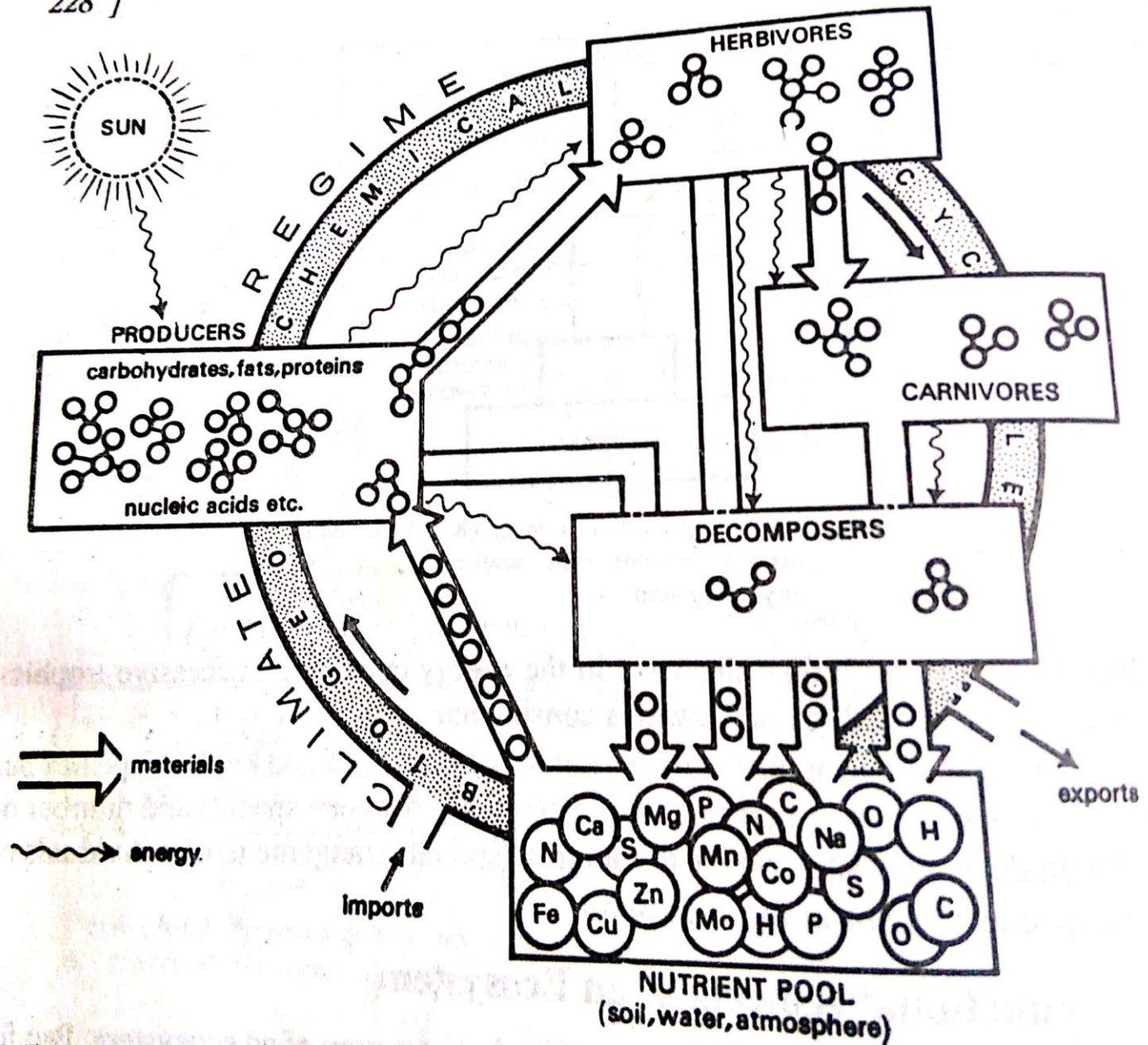
Level of Organization



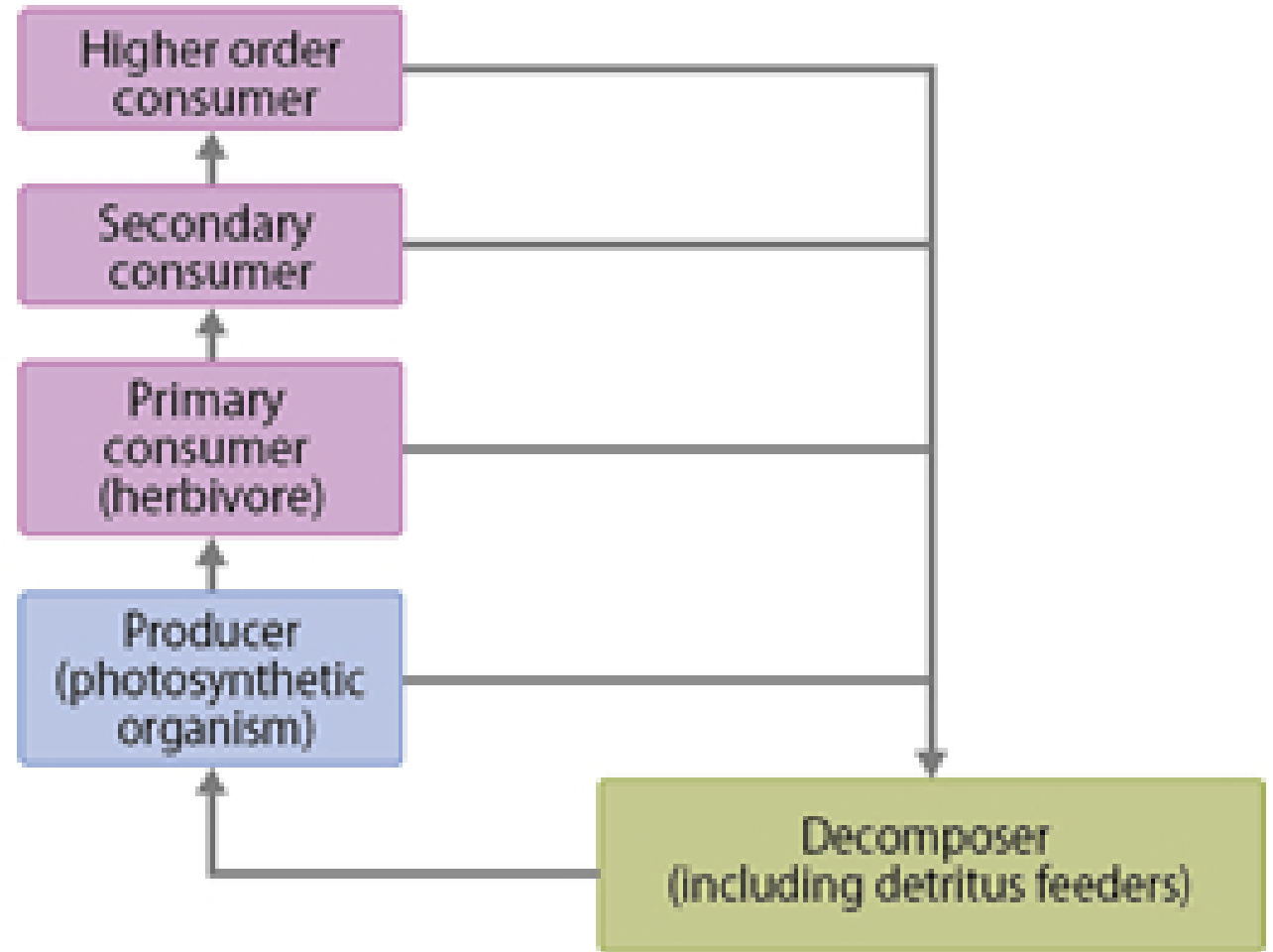
Ecosystems combine to form a Giant ecosystem



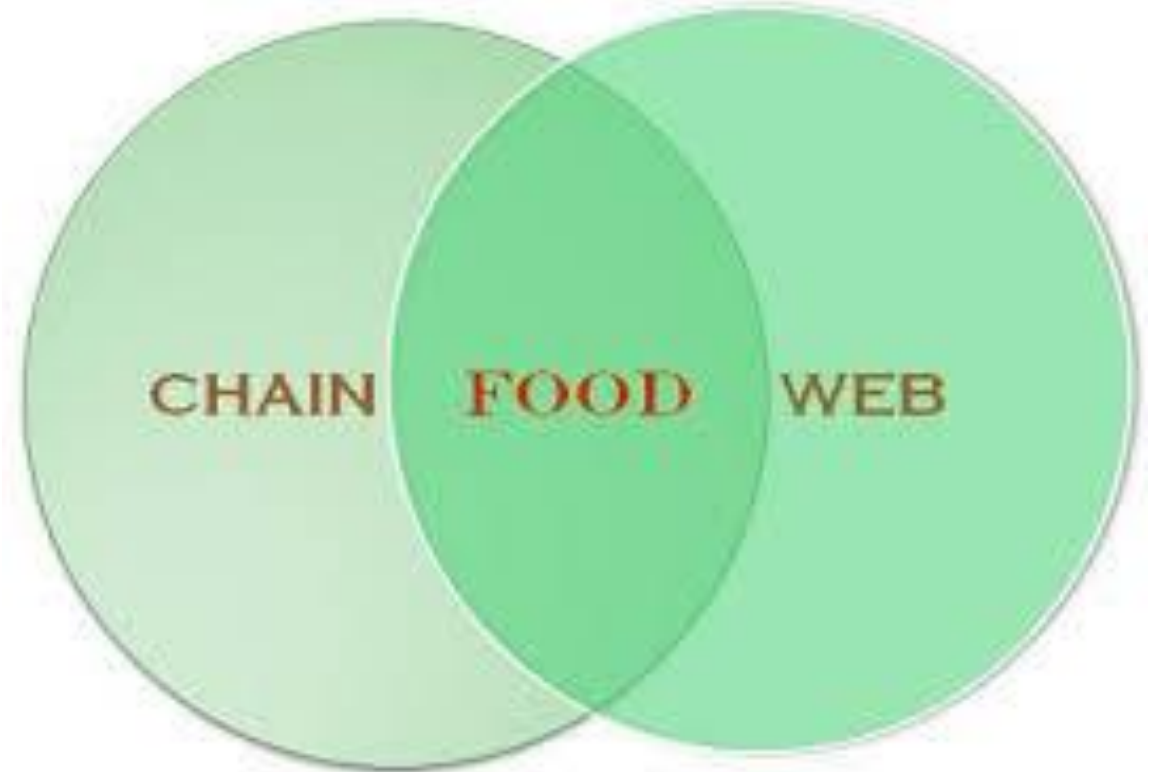
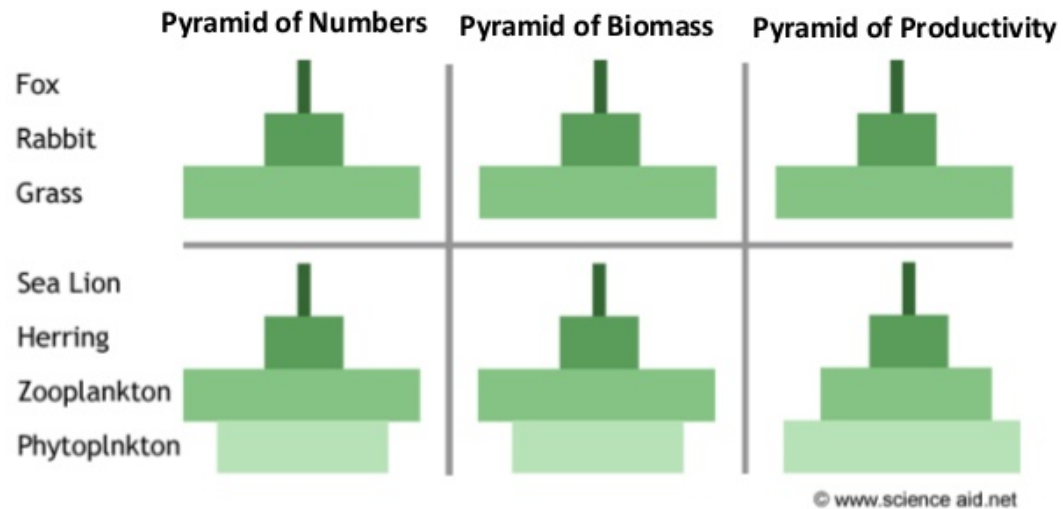
Generalized Model of Ecosystem

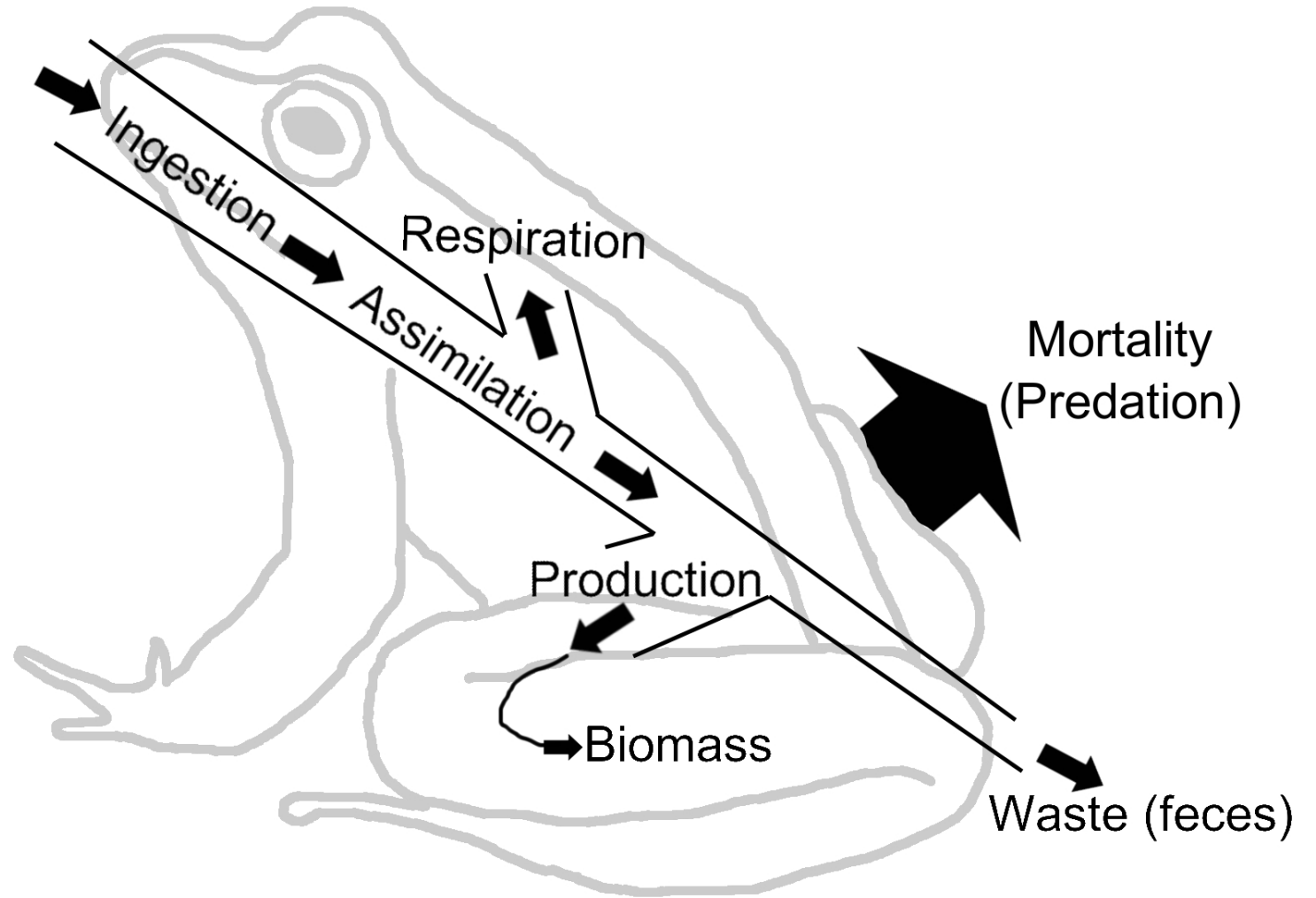


Structure and Function of Ecosystem



Ecological Pyramids and Ecological Energetics



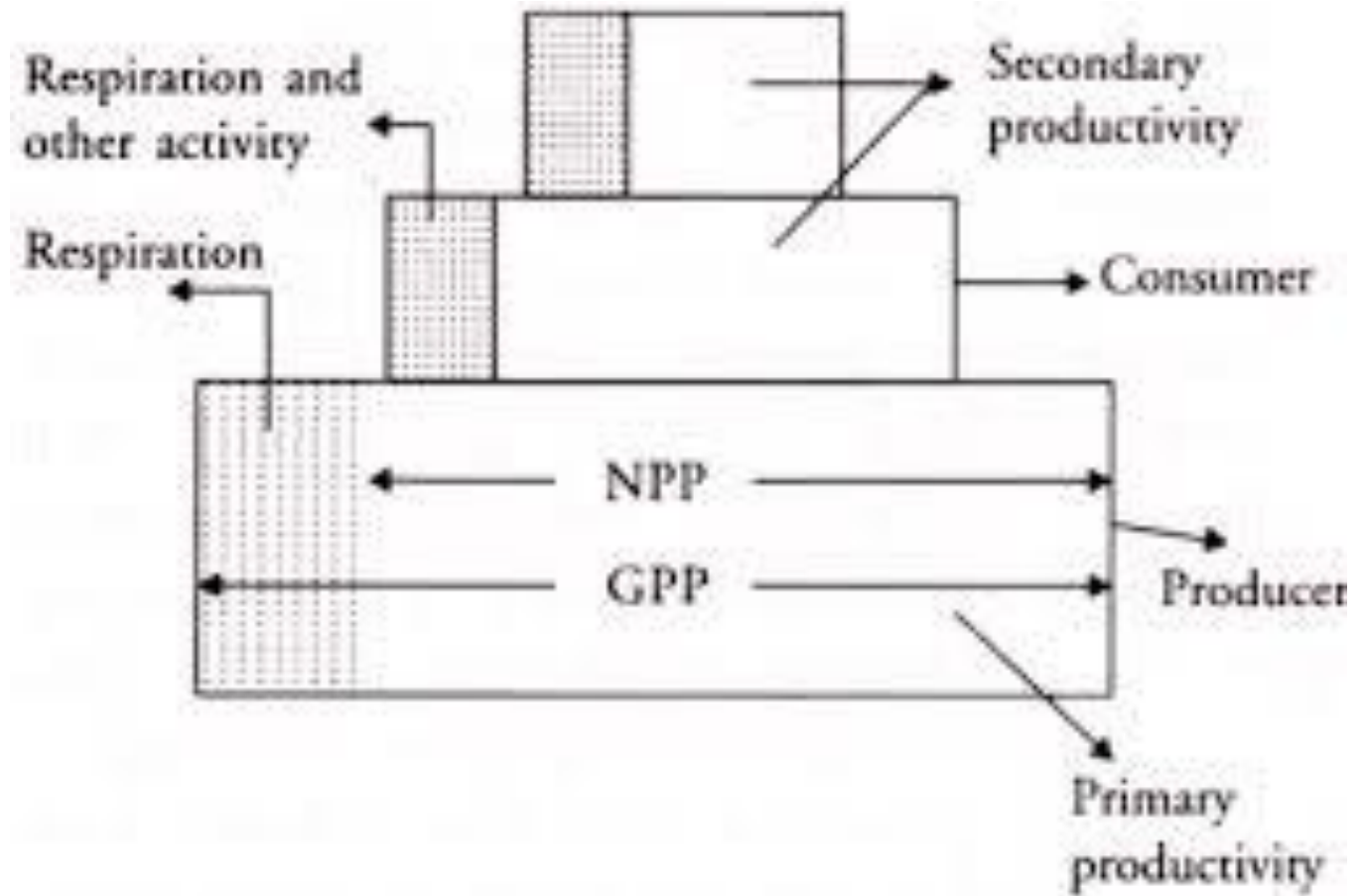


Ecological Efficiency

Ecological efficiency is a combination of several related efficiencies that describe resource utilization and the extent to which resources are converted into biomass.!

- Exploitation efficiency is the amount of food ingested divided by the amount of prey production (I_n / P_{n-1})
- Assimilation efficiency is the amount of assimilation divided by the amount of food ingestion (A_n / I_n)
- Net Production efficiency is the amount of consumer production divided by the amount of assimilation (P_n / A_n)
- Gross Production efficiency is the assimilation efficiency multiplied by the net production efficiency, which is equivalent to the amount of consumer production divided by amount of ingestion (P_n / I_n)
- Ecological efficiency is the exploitation efficiency multiplied by the assimilation efficiency multiplied by the net production efficiency, which is equivalent to the amount of consumer production divided by the amount of prey production (P_n / P_{n-1})

Ecological Productivity



Energy Flow

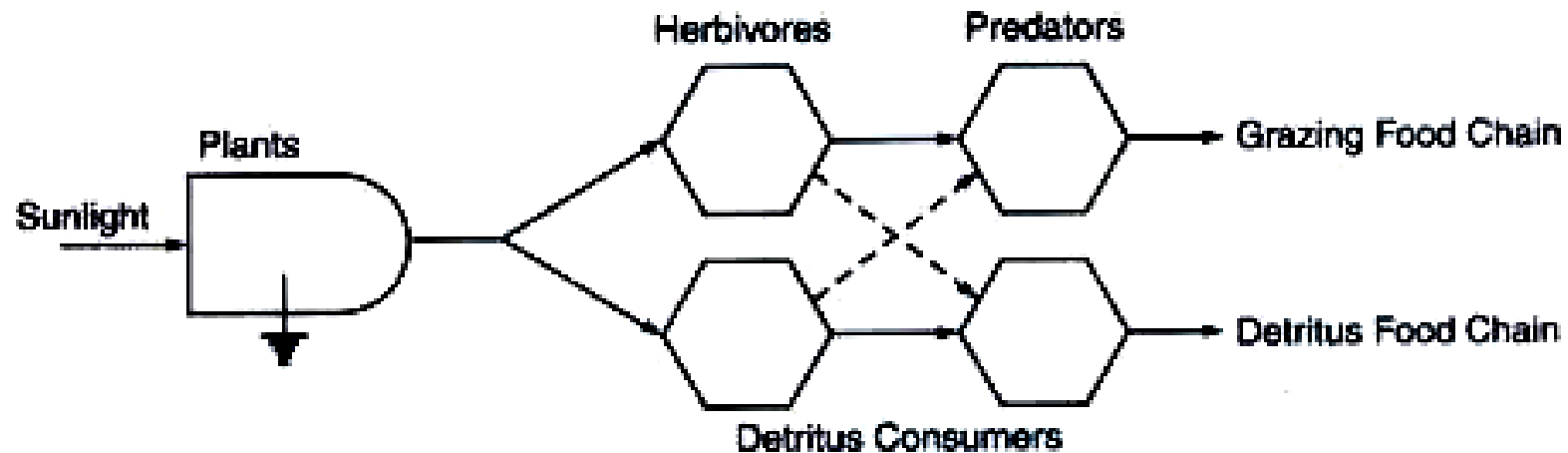
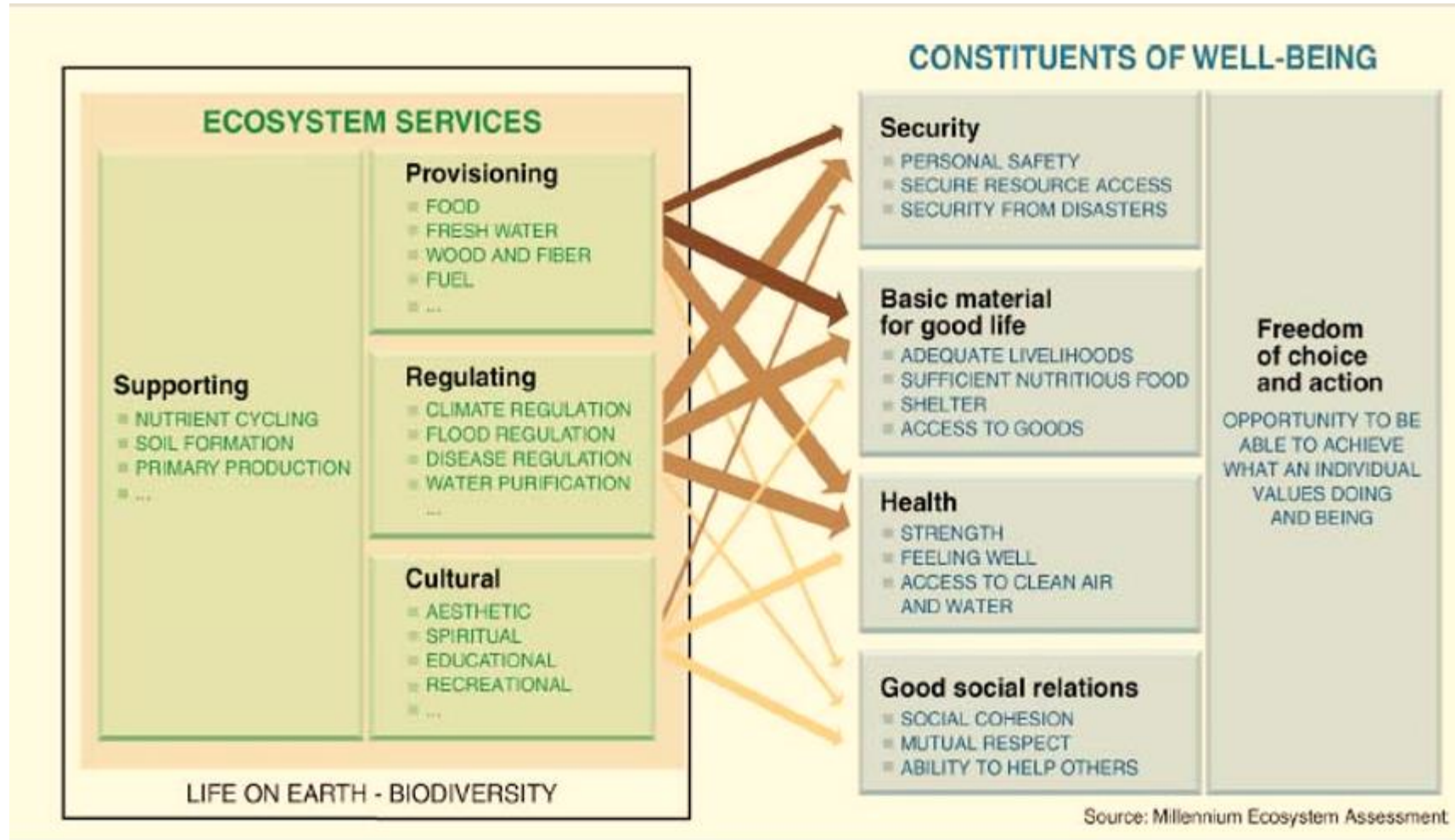


Fig. 3.2. The Y-shaped energy flow model showing linkage between the grazing and detritus food chains

Ecosystem services



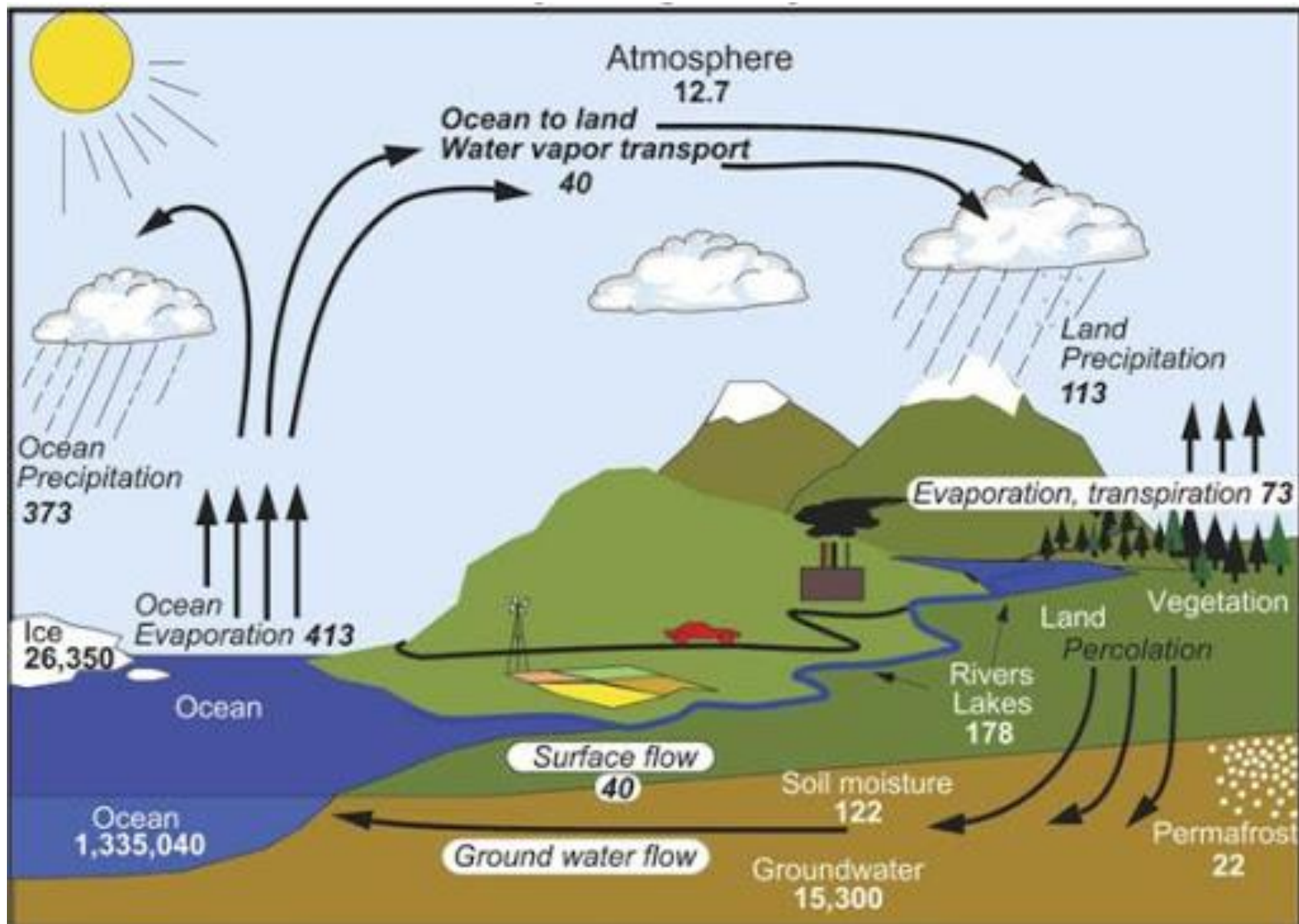
Chemical compounds for industrial and pharmaceutical uses

	Deep sea species	Function	Reference
Marketed	<i>T. thermophilus</i> enzymes - deep sea bacteria	Enzymes; Skin protection products (UV-resistant)	Arico (2005)
	<i>T. thermophilus</i> , <i>Thermus aquaticus</i> and <i>Thermatoga maritime</i> - deep sea bacteria	DNA polymerases; enzyme that builds new strands of DNA	Arico (2005)
Clinical trials	<i>Discodermia dissolute</i> - deep water sponge	Discodermolide; cancer treatment	Maxwell (2005)
	<i>Lissodenroyx sp.</i> - deep sea sponge	E7389; lung cancer and other cancer treatment	Maxwell (2005)
	<i>Salinospora tropica</i> - deep sea bacteria	Salinosporamide-1; antibiotic and anti-cancer agent	Maxwell et al (2005)
Research	<i>Lithistida</i> (family: <i>Coelistadae</i>) - deep sea sponge	Dictyostatin-1; Cancer treatment	Maxwell (2005a)
	<i>Spongosporites ruetzleri</i> - deep sea sponge	Topsentin; Anti-inflammatory agent for arthritis and skin irritations	Isbrucker et al (2003)
	<i>Isidae</i> - deep sea bamboo corals	Bone grafts	Maxwell (2005)
	<i>Vibrio diabolicus</i> - deep sea hydrothermal vent bacteria	HE 800 Exopolysaccharide; bone grafts	Zanchetta et al (2003)

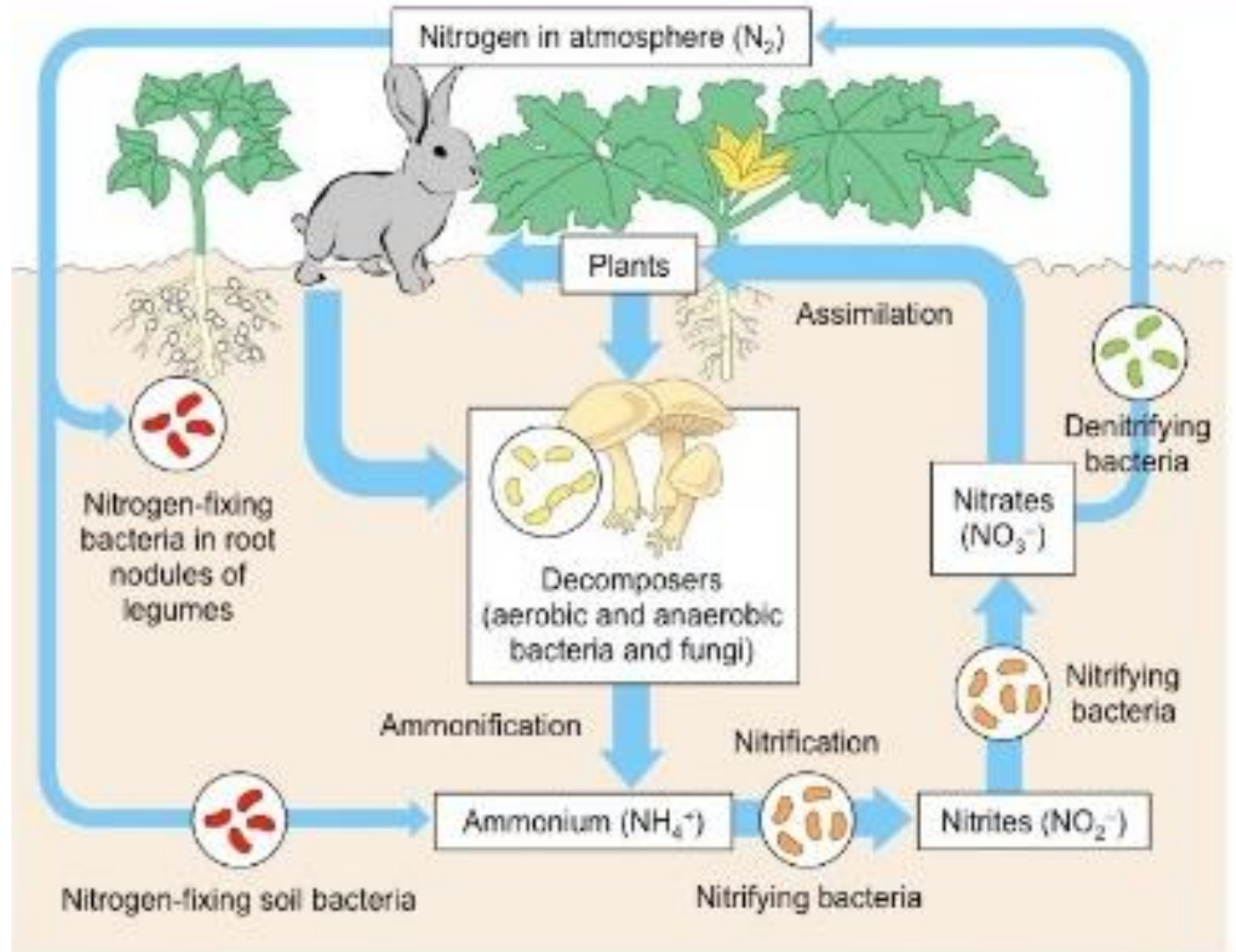
Table 2: Examples of products derived from deep-sea species and materials.

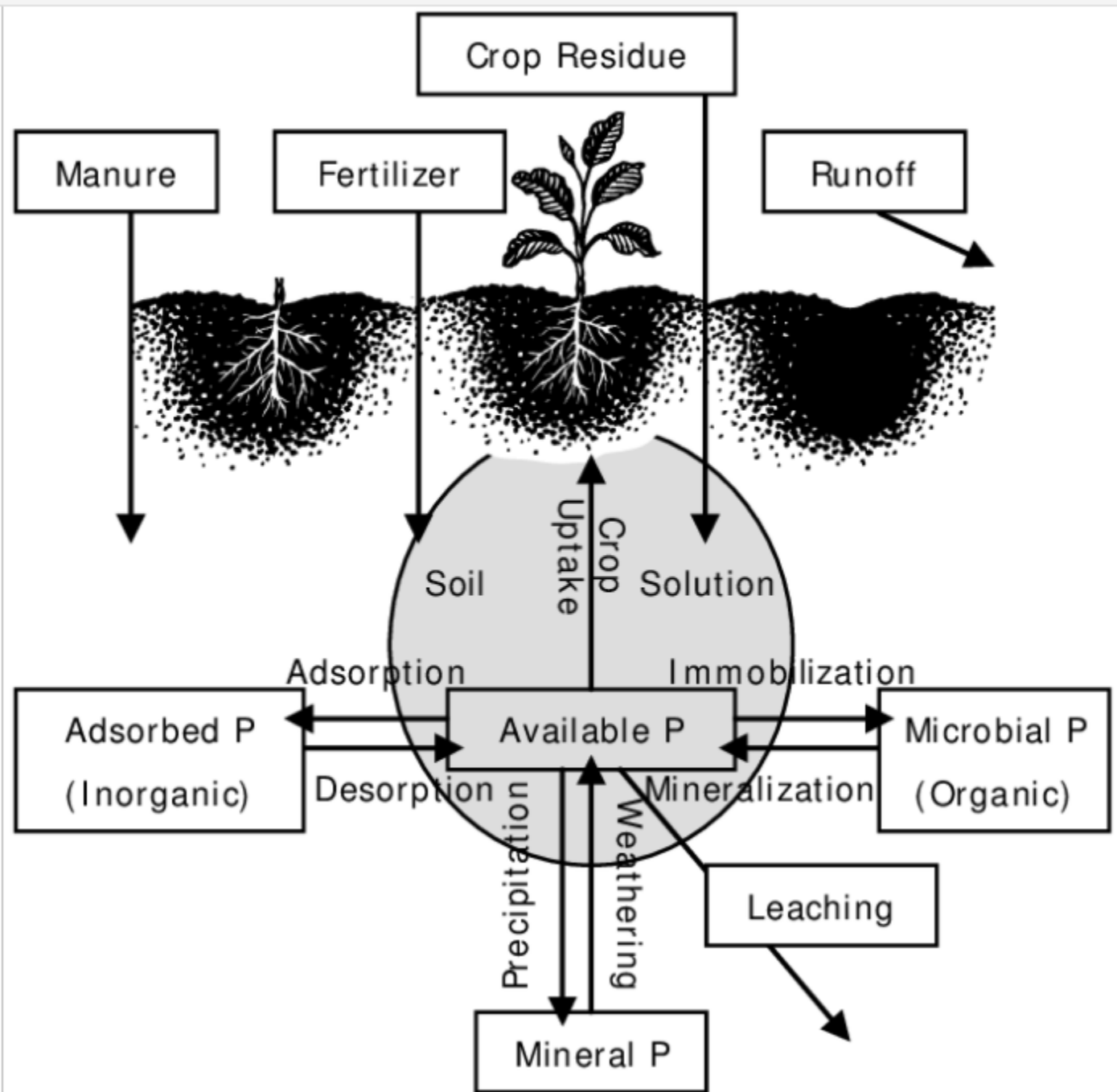
Biogeochemical cycles

- (“Bio” - living, “Geo” – earth - rock, “Chemical” - element).
- The cycling of the nutrients in the biosphere is called biogeochemical or nutrient cycle.
- It involves movement of nutrient elements through the various components of an ecosystem.
- There are more than 40 elements required for the various life processes by plants and animals; that it's proper growth and development. Most important elements are C, H, O, P, K, N, S, Ca, Fe, Mg, B, Zn, Cl, Mo, Co, I & Fe.
- These elements are continuously cycling in the ecosystem through the biogeochemical cycles and the planet earth has no input of these nutrients.
- The nutrients (matter) from the dead remains of organisms are recovered and made available to the producers by decomposers. Thus, the nutrients are never lost from the ecosystems. Exchange of nutrients between organisms and their environment is one of the essential aspects of an ecosystem.
- All organisms require nutrients for their growth, development, maintenance and reproduction.
- Circulation of nutrients within the ecosystem or biosphere is known as biogeochemical cycles and also called as 'cycling of materials'.
- There are 3 basic types, 1. Hydrologic or Water cycle, 2. Gaseous cycle (It includes atmospheric Oxygen, Carbon and Nitrogen cycles) and 3. Sedimentary cycle – It includes the cycles of Phosphorus, Sulphur and Calcium - Which are present as sediments of earth.



Nitrogen cycle consists of the following steps: fixation, ammonification, nitrification, denitrification and sedimentation.



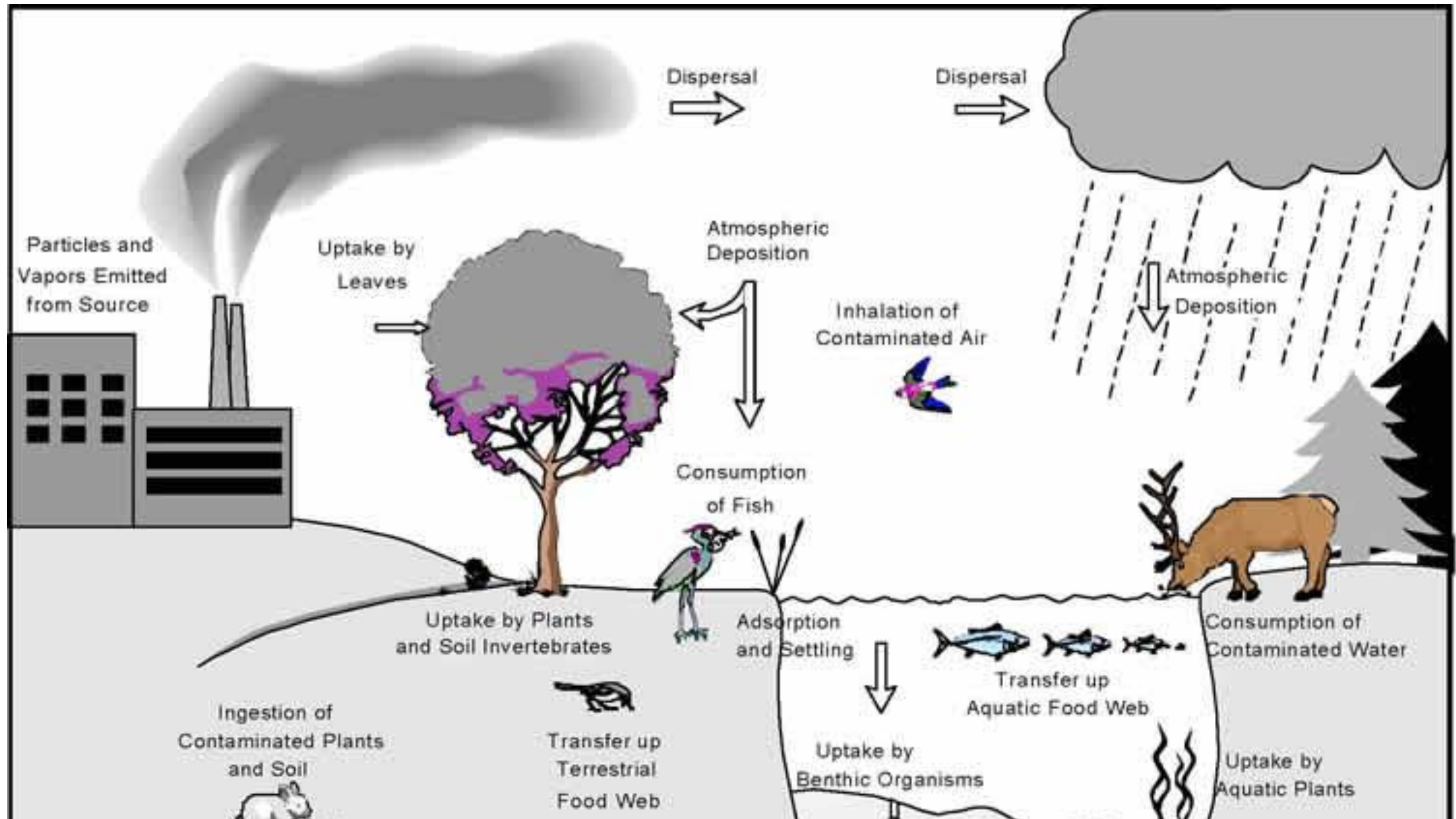


Ecosystem management

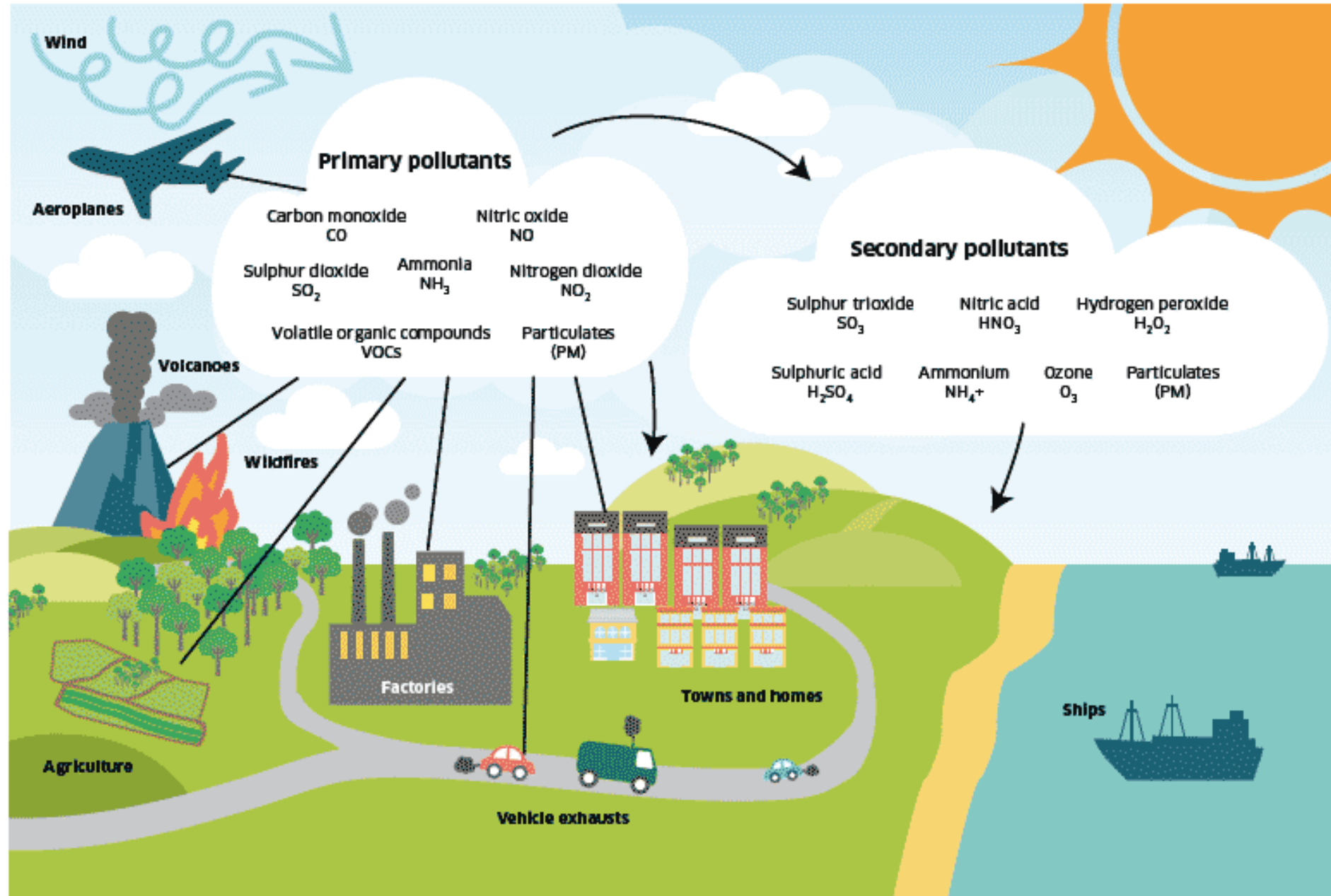
Ecosystem management is a process that aims to conserve major ecological services and restore natural resources while meeting the socioeconomic, political, and cultural needs of current and future generations.

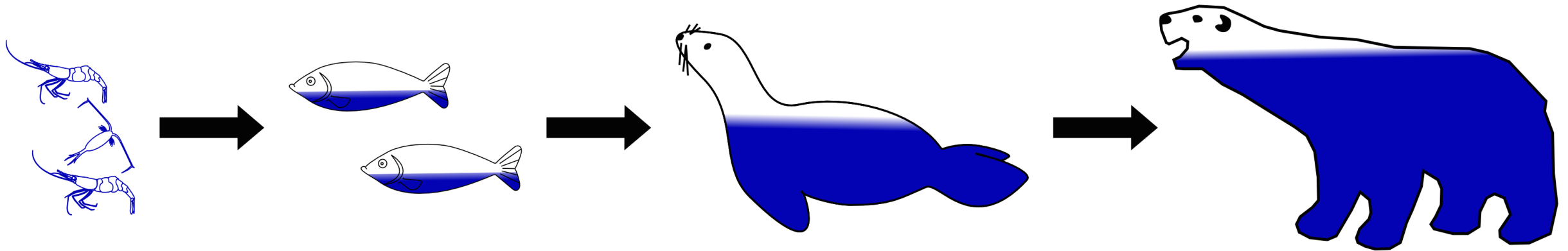
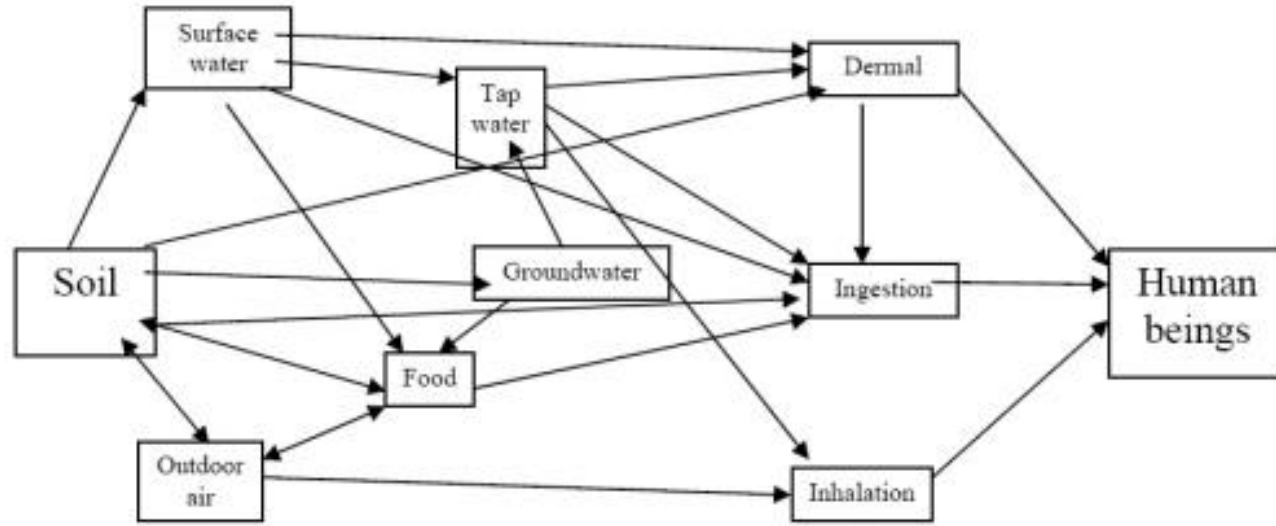
- i. adaptive management,
- ii. natural resource management,
- iii. strategic management, and
- iv. command and control management.

- Adaptive Management is based on future predictions
- Natural resource management refers to the management of natural resources such as land, water, soil, plants and animals, with a particular focus on how management affects the quality of life for both present and future generations
- Strategic management encourages the establishment of goals that will benefit the ecosystem while keeping socioeconomic and politically relevant issues in mind
- Command and control management utilizes a linear problem-solving approach where a perceived problem is solved through controlling devices such as laws, contracts and/or agreements



Fate of environmental pollutants





 Contaminant Levels

Biomagnification

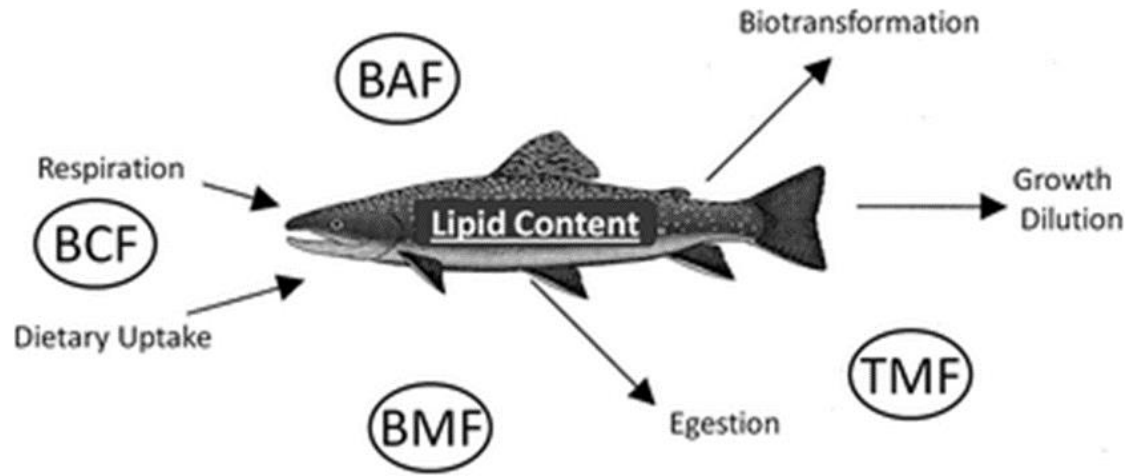


Table 1 Definitions for BCF, BAF, BMF and TMF

Bio-uptake factors	Definition
Bioconcentration factor (BCF)	Ratio of fish to water concentrations with no dietary intake
Bioaccumulation factor (BAF)	Ratio of fish to water concentrations with dietary intake
Biomagnification factor (BMF)	Ratio of fish to diet concentrations
Trophic magnification factor (TMF)	Averaged BMF over a food web of several trophic levels

1. The concentration of Hg in a waterbody is 0.55 mg/L. However, concentration in small fishes were found as 0.8 mg/Kg and in large fish as 1.56 mg/Kg. Calculate the BCF, BAF, BMF.

2. In a locality the soil was found to be contaminate with 0.45mg/Kg of Cd. A number of wild herbs were collected from the locality and average Cd concentration was found to be like this

Species A- 0.56 mg/Kg

Species B – 0.25 mg/Kg

Speies C – 1.22 mg.Kg

a. What are their BCFs? Which species are hyperaccumulators?

In the same locality 1st level consumer X and Y feeding on A and C respectively were found to contain 2.8 mg/Kg and 2.6 mg/Kg Cd .

b. What were their BAF and BMF? What was the average TMF?



SPECIAL REPORT
NIC 2009-03D April 2009

India: Impact of Climate Change to 2030 A Commissioned Research Report

Prepared By Joint Global Change Research Institute and
Battelle Memorial Institute, Pacific Northwest Division

THE NATIONAL INTELLIGENCE COUNCIL SPONSORS WORKSHOPS AND RESEARCH WITH
NONGOVERNMENTAL EXPERTS TO GAIN KNOWLEDGE AND INSIGHT AND TO SHARPEN DEBATE
ON CRITICAL ISSUES. THE VIEWS EXPRESSED IN THIS REPORT DO NOT REFLECT OFFICIAL
US GOVERNMENT POSITIONS.

- ❑ India is both a major greenhouse gas emitter and one of the most vulnerable countries in the world to projected climate change.
- ❑ The country is already experiencing changes in climate and the impacts of climate change, including water stress, heat waves and drought, severe storms and flooding, and associated negative consequences on health and livelihoods.
- ❑ With a 1.3 billion but growing population and dependence on agriculture, India probably will be severely impacted by continuing climate change.

Global observations of melting glaciers suggest that climate change is well under way in the region, with **glaciers receding at an average rate of 10–15 meters per year**. If the rate increases, **flooding is likely** in river valleys fed by these glaciers, **followed by diminished flows**, resulting in water **scarcity for drinking and irrigation**.

All models show a trend of general warming in mean annual temperature as well as decreased range of diurnal temperature and enhanced precipitation over the Indian subcontinent. **A warming of 0.5° C is likely over all India by the year 2030 (approximately equal to the warming over the 20th century) and a warming of 2-4° C by the end of this century**, with the maximum increase over northern India. Increased warming is likely to lead to higher levels of tropospheric ozone pollution and other air pollution in the major cities.

Increased precipitation—including monsoonal rains—is likely to come in the form of **fewer rainy days but more days of extreme rainfall events**, with increasing amounts of rain in each event, leading to significant **flooding**. **Drizzle-type precipitation that replenishes soil moisture is likely to decrease**. The timing may also shift, causing a drying during the late summer growing season.

Specific effects

- Agriculture. High-input, high-output agriculture will be negatively affected even as demands for food and other agricultural products rise because of an increasing population and expectations for an improved standard of living.
- Water: Glacier melt may yield more runoff in the short term but less in the medium and long terms. More severe storms (especially cyclones) will cause more damage to infrastructure and livelihoods and exacerbate salt water intrusion in storm surges.
- Changes in the timing and amount of monsoon rains will make the production of food and other agricultural products more uncertain, so that, even in good-weather years, farmers will be more likely to make decisions leading to lower-productivity.

- **Exacerbation of Inequality:** The welfare of those who are affected by climate change and who have limited means to adapt may act as a force that can change governments, strain public budgets, and foster unrest. About one-third of Indians are extremely poor, and 60 percent depend upon agriculture for their livelihoods.
- **Energy:** As India searches for additional sources of energy to meet rising demand, climate change mitigation efforts may constrain its use of indigenous and imported coal, oil, and gas, while development of nuclear energy will be slow at best and likely to encounter opposition. Other non-emitting technologies will require technology transfer and capacity-building.
- **Migration:** India receives immigrants from a number of countries. Under climate change conditions, it may be flooded with many more, particularly from Bangladesh. Such migration may exacerbate tension between the two countries as well as putting a strain on Indian central and state governments.